Abdul Basith Maaz

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Professional Summary

Backend & GenAI Engineer with production ownership across 5 SaaS products, including a high-concurrency FastAPI platform serving 50K+ users. Specialized in Tri-Tier Multi-Agent Architectures (CrewAI, AutoGPT, LangGraph), Hybrid Vector Retrieval (pgvector RRF), and High-Throughput Production Systems with zero-knowledge local privacy. Built and launched the Waddi CRM solo from ideation to production.

Experience

Bottrion Systems Pvt. Ltd. Remote
Software Development Engineer Aug 2026 – Present
Junior Full Stack Engineer Aug 2025 – July 2026

- Took sole backend ownership of Khayyal, a Saudi SaaS platform serving 50K+ users; reverse-engineered and maintained its FastAPI microservice architecture with zero downtime during senior transition.
- Architected and launched Waddi CRM solo using Django REST Framework, React, TypeScript, PostgreSQL, and Celery, onboarding 100% of target sales operations and cutting $1,200/month in third-party tooling.
- Eliminated checkout state race conditions across Razorpay, STC Pay, and Dhamen Pay causing 25+ false-negative orders/week by implementing row-level database locks (`select_for_update()`) and idempotent Redis keys.
- Led zero-downtime migration of 131 endpoints from FastAPI to Django with API versioning (`/api/v1/` to `/api/v2/`) and Redis schema caching, reducing OpenAPI documentation load from 10s to under 850ms.
- Owned GCP cloud infrastructure, Docker containerization, and CI/CD pipelines across 5 SaaS products; standardized Linux telemetry workflows, cutting mean-time-to-recovery (MTTR) from hours to minutes.

IBM & Rooman Technologies Pvt. Ltd. Sep 2024 – Mar 2025
AI / DevOps Intern

- Implemented Jenkins CI/CD automation pipelines and isolated Docker testing environments for a 4-person engineering team, reducing staging deployment setup overhead by 40%.

Technical Skills

Multi-Agent Systems: CrewAI, AutoGPT, LangGraph, LlamaIndex Synthesizers
Retrieval & Vector DBs: PostgreSQL pgvector (HNSW & IVFFlat), Reciprocal Rank Fusion, Hybrid Search, Semantic Caching, FastEmbed ONNX
LLMs & Guardrails: Gemini 2.5 Flash / Flash-Lite, Prompt Injection Defense Sandboxes, Thin-Evidence Gating, Pydantic Structured Outputs, Token Budget Management
Backend & Cloud: Python, FastAPI, Django REST Framework, Celery, Redis, Docker,Linux, GCP, Nginx, RESTful APIs, Git
Languages & Web: Python, TypeScript, JavaScript, SQL, React.js, Next.js, HTML5/CSS3

Projects

Shelf | Live Product (theonlyshelf.vercel.app) *Next.js 15, FastAPI, CrewAI, AutoGPT, pgvector, LangGraph*

- Architected a Tri-Tier Agentic Hierarchy combining LangGraph (sub-second deterministic search gating), CrewAI (collaborative 3-agent expense auditing and SRE incident triage swarms), and an AutoGPT-style autonomous ReAct loop with mandatory Self-Criticism, eliminating runaway execution loops via strict 5-step safety bounds.
- Implemented Reciprocal Rank Fusion (RRF, $k = 60$) in PostgreSQL pgvector over HNSW indexes, fusing sparse OCR full-text, dense bge-small text vectors (384d), and CLIP vision embeddings (512d), elevating Recall@5 from 0.833 to 1.000 and Mean Reciprocal Rank (MRR) from 0.722 to 0.903 on benchmark evaluations.
- Engineered a dual-engine ingestion pipeline utilizing client-side WebAssembly OCR (Tesseract.js WASM at ∼600ms latency) for zero-knowledge data privacy, paired with a concurrent serverless worker pool extracting scene captions and 15+ automated semantic tags.

AI Mock Interview Platform *Next.js, MediaPipe WASM, Whisper, Parakeet, Gemini 2.5 Flash*

- Re-architected facial landmark telemetry from server GPU queues to client-side MediaPipe WebAssembly (468 points), reducing latency from 60s to 5.2s, saving $400+/month in compute, and enforcing zero-knowledge privacy.
- Engineered dual audio transcription pipeline fusing Whisper and Parakeet with Redis caching, generating structured executive behavioral and emotion entropy assessments with Gemini 2.5 Flash.

Wakat *Django, Celery, React, PostgreSQL, ECDSA secp256k1*

- Shipped a multi-tenant ZATCA Phase 2 e-invoicing engine for Saudi enterprises: UBL 2.1 XML invoices with sequential per-device SHA-256 hash chains, ECDSA digital signatures, and 9-tag binary TLV QR codes cleared via Fatoora API.
- Built the core financial backend in Django and Celery handling double-entry journals, chart of accounts, automated VAT returns, and cryptographic CSID device onboarding lifecycle.

Education

Bachelor of Engineering in Information Science and Engineering 2021 – 2025
Visvesvaraya Technological University (VTU) Karnataka, India

Publication & Certifications

Publication: Advancements in Deep Learning for Autonomous Driving in Indian Road Conditions, *Journal of Intelligent Data Analysis*, 2024.
Certifications: NASSCOM AI DevOps Engineer • NPTEL Cloud Computing • IBM Introduction to Cloud • IBM DevOps Practices